Laurent “Enzo” Ntibarikure

R&D/Product Development Engineer | Electronics & Packaging Specialist | AI driven Automation

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Summary

Principal R&D/Product Development Engineer with 15+ years experience in signal integrity design, simulation, and validation for high-speed electronic systems. Expert in establishing signal integrity requirements, performing high-speed simulations, and driving product development from concept through production. Proven track record in connector design, electronic packaging, and high-speed differential signaling applications.

Experience

Sr. Engineer - Design Enablement (Electromagnetic Flow Developer and Modelling Engineer, Grade G3)

NXP Semiconductors | *03/2022 – Present* | Eindhoven, Netherlands

- Developed Ansys Electronics Design Synchronization Plugin, reducing cost of ownership and improving version control traceability (4/5 customer satisfaction from RFP stakeholders)
- Benchmarked and verified MMICs with Cadence EMX and Keysight RFPro
- Delivered Ansys Electronics benchmarking, technical support, and knowledge-sharing presentations
- Performed signal integrity analysis and validation for SoCs and antenna systems; established SI requirements for customer applications
- Built automation workflows cutting manual effort by 30%, boosting signal/power integrity analysis efficiency for PCB/package/advanced packaging
- Delivered HFSS/SIwave consulting and training as signal integrity expert for product development teams
- Collaborated with global design teams across US, Europe and Asia to deliver signal integrity solutions

Technical Account Manager

Ansys | *02/2020 – 03/2022* | Utrecht, Netherlands

- Served as signal integrity expert (HFSS, SIwave, Q3D Extractor) supporting SI analysis and validation for product development projects
- Led cross-functional teams architecting integrated SI validation flows for electronic packaging and high-speed interconnect systems

- Advised on SI requirements and test methodologies, influencing $1M+ in annual opportunities
- Delivered HFSS/SIwave/Q3D Extractor consulting and training for customers and internal teams
- Built RF/signal/power integrity/EMC automation workflows improving support efficiency and analysis throughput
- Managed $500K annual technical budgets and signal integrity analysis teams (5-8 members)

Sr. Applications Engineer

Ansys | *01/2019 – 03/2020* | Utrecht, Netherlands

- Provided signal integrity expertise throughout the product development lifecycle; led SI requirements gathering and validation scoping for high-speed applications
- Led SI workshops on automotive electrification and connectivity, delivering best-in-class SI/PI validation approaches
- Ran HFSS, SIwave, Q3D Extractor high-frequency simulations for electronic packaging and high-speed interconnect systems
- Collaborated with product teams to define SI requirements and validate innovative features
- Developed standardized SI analysis frameworks cutting project initiation time by 25%

Lead EIC Engineer - Testing

Baker Hughes | *07/2016 – 12/2018* | Florence, Italy

- Led development of ATEX-compliant distributed data acquisition systems for electronic testing applications
- Resolved 5+ EMI issues in new product introductions and factory acceptance tests through electromagnetic analysis
- Received innovation award for Structural Vibration Monitoring System patent
- Championed automation cutting manual design of test procedures by 40%, improving testing efficiency and repeatability

EIC Engineer - Testing

Baker Hughes | *06/2013 – 06/2016* | Florence, Italy

- Executed electrical, instrumentation and controls design for new product introductions
- Contributed to rapid test executions through standardized procedures and automation
- Assisted in project coordination for test system deployments supporting product validation
- Applied high-speed test equipment principles to signal integrity validation

Startup Co-Founder

Università degli Studi di Firenze | *06/2012 – 06/2013* | Florence, Italy

- Co-founded ElectroSoft, commercializing efficient electromagnetic numerical codes for simulation applications

- Led business development including market analysis, technical feasibility studies, and investor presentations
- Identified defense industry opportunities through technical demonstrations and proof-of-concept work
- Applied electromagnetic theory and numerical methods to develop efficient simulation algorithms

Projects

GE Aviation's Catalyst

Baker Hughes, a GE Company | *06/2017 - 12/2018* | Prague, Czech Republic

- Led special instrumentation design for FETT, contributing to >$4M project

GE Power slip-ring digital telemetry

Baker Hughes, a GE Company | *04/2017 - 05/2017* | Florence, Italy

- Ran Ansys HFSS simulations optimizing signal/power integrity over slip-ring connections
- Achieved ring spacing reduction lowering manufacturing costs

GE-6G PowerGen Module Structural Vibrations Monitoring System

GE Oil & Gas | *11/2015 - 03/2016* | Massa, Italy

- Developed ATEX-compliant distributed acquisition system for 150+ vibration points
- Introduced software velocity/displacement computation saving $150k directly

Power density centrifugal compressor

GE Oil & Gas | *03/2014 - 05/2016* | Florence, Italy

- Led Fieldbus Foundation commissioning for 240+ pressure transducers (0.1% uncertainty)
- Managed ATEX-compliant capacitive vibration probes
- Developed electrostatic numerical model for capacitance prediction

Finite Element Software (FES)

Università degli Studi di Firenze | *06/2011 - 02/2014* | Florence, Italy

- Developed multi-backend (C++, Python, MATLAB) open-source FEA solver for EM/thermal
- Implemented core formulations: 3D frequency-domain, 2D TMz, electrostatic, waveport eigenmode
- Integrated advanced features: TFE method, domain decomposition, nonlinear Kerr materials
- Verified against HFSS/CST meshes; developed automated testing frameworks

Phased array antennas model order reduction (MEng thesis)

Universität des Saarlandes | *10/2009 - 02/2010* | Saarbrücken, Germany

- Developed intrusive model order reduction for fast, accurate antenna array beamforming

- Implemented Matlab code achieving 3000x matrix reduction and 300x speedup

RS-232 modem in ISM 868 MHz band (BEng thesis)

University of Florence | *01/2007 - 04/2007* | Florence, Italy

- Created wireless connectivity for avalanche rescue radar, optimizing firmware timing
- Designed and assembled ISM 868 MHz modem PCB with off-the-shelf MCU and Radio

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Education

PhD, Electronics Engineering | Università degli Studi di Firenze | *01/2011 – 02/2014*

- Thesis: “Contributions to the Art of Finite Element Analysis in Electromagnetics”
- Advisors: Prof. Giuseppe Pelosi, Dr. Stefano Selleri
- School of RF, Microwaves and Electromagnetics; Computational Methods focus

Master’s, Electronics Engineering | Università degli Studi di Firenze | *10/2007 – 04/2010*

- Thesis: “Model Order Reduction in Full-Wave Analysis of Phased Array Antennas”
- Advisors: Prof. Giuseppe Pelosi, Dr. Ing. Stefano Selleri; External: Prof. Dr. Romanus Dyczij-Edlinger, Dr.-Ing. Ortwin Farle
- Erasmus Placement at Lehrstuhl für Theoretische Elektrotechnik, Universität des Saarlandes

Bachelor’s, Electronics Engineering | Università degli Studi di Firenze | *01/2003 – 04/2008* **High School Diploma** | École Autonome Belge à Bujumbura | *09/1996 – 06/2002* | Bujumbura, Burundi

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Key Certifications

- Professional Engineer, ID: ES2010277049000038 — Università degli Studi di Firenze
- IEEE Member ID:90681720, over 13+ years
- Ansys Certification: Getting Started with Cloud (Pre-Sales Support 3050)
- NXP Course: Strategies for Negotiation (Completed Jul 2023)

Selected Courses

- Command of the Message — Ansys, 06/2020
- JAWS: Just Another Way of Selling — Ansys, 02/2020
- GE Crotonville: Delivering Customer Impact — GE Oil & Gas, 03/2016
- CE Marking Overview — GE Oil & Gas, 11/2014
- Safety Integrity Levels — GE Oil & Gas, 10/2014

- Total Quality Lean Six Sigma Green Belt — GE Oil & Gas, 06/2014

Technical Skills

- Signal Integrity Analysis: Ansys HFSS, SIwave, Q3D Extractor, CST Studio Suite, Agilent ADS
- PCB Design & Layout: Altium Designer, AutoCAD Electrical, Mentor Graphics
- 3D CAD & Modeling: SpaceClaim, AutoCAD 3D, Creo Parametric
- High-Speed Test Equipment: VNA, TDR, BERT, Oscilloscopes
- Electromagnetic Simulation & Testing; Finite Element Analysis (FEA); RF/Microwave & Signal/Power Integrity Analysis
- Scripting: Matlab, C++, C, Python; Data Analysis & Visualization

Professional Skills

- Signal Integrity Leadership & Mentoring; Technical Presentation & Paper Authorship
- Competitive Analysis & IP Strategy; Project Leadership & Cross-Functional Team Management
- Product Development Lifecycle Management; Requirements Definition & Validation
- Design for Manufacturing & Assembly; Process Improvement & Innovation
- Customer Technical Engagement & Support; Documentation & Specification Development
- Problem-Solving & Analytical Thinking; Business Acumen & Strategic Growth

Languages

English (Native/Bilingual) · French (Native/Bilingual) · Italian (Native/Bilingual) · Kirundi (Professional) · Spanish (Elementary)

Publications Highlights

- “Multiphysics simulations for 5G RFICs and SoCs.” *Bits&Chips*, 2019.
- “Fiber-Optic Liquid Level Sensing by Temperature Profiling with an FBG Array.” *Sensors*, 2018.
- “Harmonic Balance Domain Decomposition Finite Elements for Nonlinear Passive Microwave Devices Analysis.” *Electromagnetics*, 2014.
- “Efficient Harmonic Balance Analysis of Waveguide Devices With Nonlinear Dielectrics.” *IEEE Microwave and Wireless Components Letters*, 2012.
- Book: *Contributions to the Art of Finite Elements in Electromagnetics* (2014)
- Book Chapter: “Sensoristica in ambiente estremo: l’interno di una turbina Baker Hughes” (2026)
- Multiple conference proceedings on GMRES, harmonic balance, and model order reduction
- Patent: “Module with Sensors Arrangement” US20180180464A1 (2018)